

2-D Kinematics

SS ILP Physics (Intermediate), Week 1 Class 4



Outline

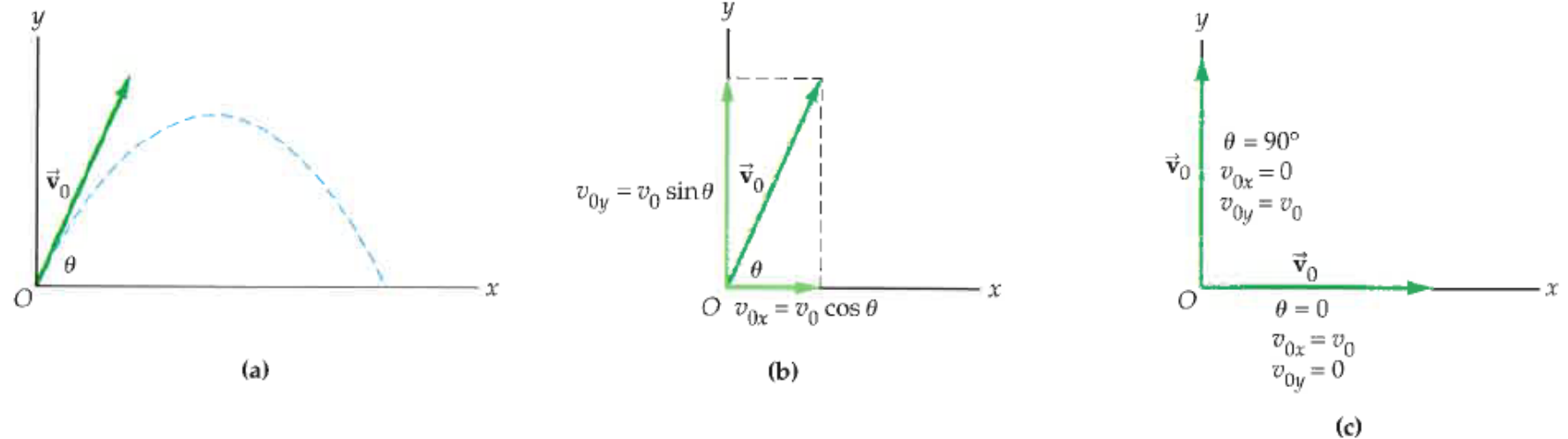
- Projectile Motion: General launch angle
- Parabolic trajectory
- Symmetry in projectile motion
- Maximum height

Projectile Motion: General launch angle

When a projectile is launched with an initial velocity $\vec{v}_0$ at an arbitrary angle θ with respect to the horizontal,

$$v_{x,0} = v_0 \cos \theta$$

$$v_{y,0} = v_0 \sin \theta$$



▲ **FIGURE 4-6** Projectile with an arbitrary launch angle

Example

A ball is thrown out with an initial velocity of 300 [m/s] at an angle of 60° measured from the horizontal. The ball landed on a mountainside 40s later. Take $g = 9.81 \text{ m/s}^2$.

- a) What are the horizontal and vertical components of the initial velocity?
- b) What are the horizontal and vertical displacement of the ball at time $t = 40\text{s}$?

<http://physicshelpline.net/files/documents/17.01.12.pdf>

Example: Solution

A ball is thrown out with an initial velocity of 300 [m/s] at an angle of 60° measured from the horizontal. The ball landed on a mountainside 40s later. Take $g = 9.81 \text{ m/s}^2$.

- a) What are the horizontal and vertical components of the initial velocity?
- b) What are the horizontal and vertical displacement of the ball at time $t = 40\text{s}$?

a)

$$v_{x,0} = 300 \cos 60^\circ = 150 \text{ m/s}$$
$$v_{y,0} = 300 \sin 60^\circ = 260 \text{ m/s}$$

b) At time $t = 40 \text{ s}$,

$$x = x_0 + v_{x,0}t = 0 + 150(40) = 6000 \text{ m}$$
$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 = 0 + 260(40) + \frac{1}{2}(-9.81)(40)^2 = 2552 \text{ m}$$

<http://physicshelpline.net/files/documents/17.01.12.pdf>

Case Problem 1

A projectile is launched from the origin with an initial speed of 20.0 m/s at an angle of 35.0° above the horizontal. Take $g = 9.81 \text{ m/s}^2$. Find the x and y positions of the projectile at times

a) $t = 0.5 \text{ s}$

b) $t = 1.0 \text{ s}$

Case Problem 1: Solution

$$\text{a) } v_{x,0} = v_0 \cos \theta = 20 \times \cos 35.0^\circ = 16.38 \text{ m/s}$$

$$v_{y,0} = v_0 \sin \theta = 20 \times \sin 35.0^\circ = 11.47 \text{ m/s}$$

$$x = x_0 + v_{x,0}t = 16.38 \times 0.5 = 8.19 \text{ m}$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 = 11.47 \times 0.5 - \frac{1}{2}g \times (0.5)^2 = 4.51 \text{ m}$$

Case Problem 1: Solution

$$\text{b)} \quad v_{x,0} = v_0 \cos \theta = 20 \times \cos 35.0^\circ = 16.38 \text{ m/s}$$

$$v_{y,0} = v_0 \sin \theta = 20 \times \sin 35.0^\circ = 11.47 \text{ m/s}$$

$$x = x_0 + v_{x,0}t = 16.38 \times 1.0 = 16.38 \text{ m}$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 = 11.47 \times 1.0 - \frac{1}{2}g \times (1.0)^2 = 6.57 \text{ m}$$

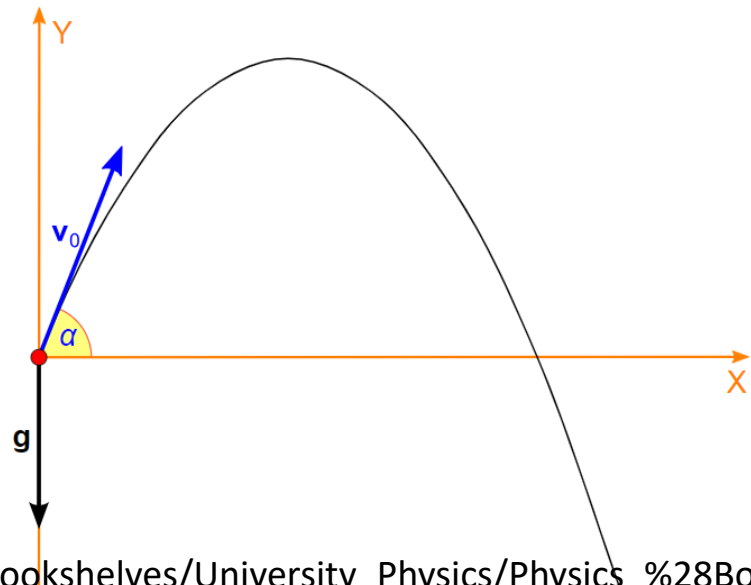
Parabolic trajectory

The shape of the trajectory followed by a projectile is parabolic.

$$x = v_{x,0}t \rightarrow t = \frac{x}{v_{x,0}}$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 = y_0 + v_{y,0}\left(\frac{x}{v_{x,0}}\right) - \frac{1}{2}g\left(\frac{x}{v_{x,0}}\right)^2$$

$$y = y_0 + \left(\frac{v_{y,0}}{v_{x,0}}\right)x - \frac{g}{2(v_{x,0})^2}x^2$$

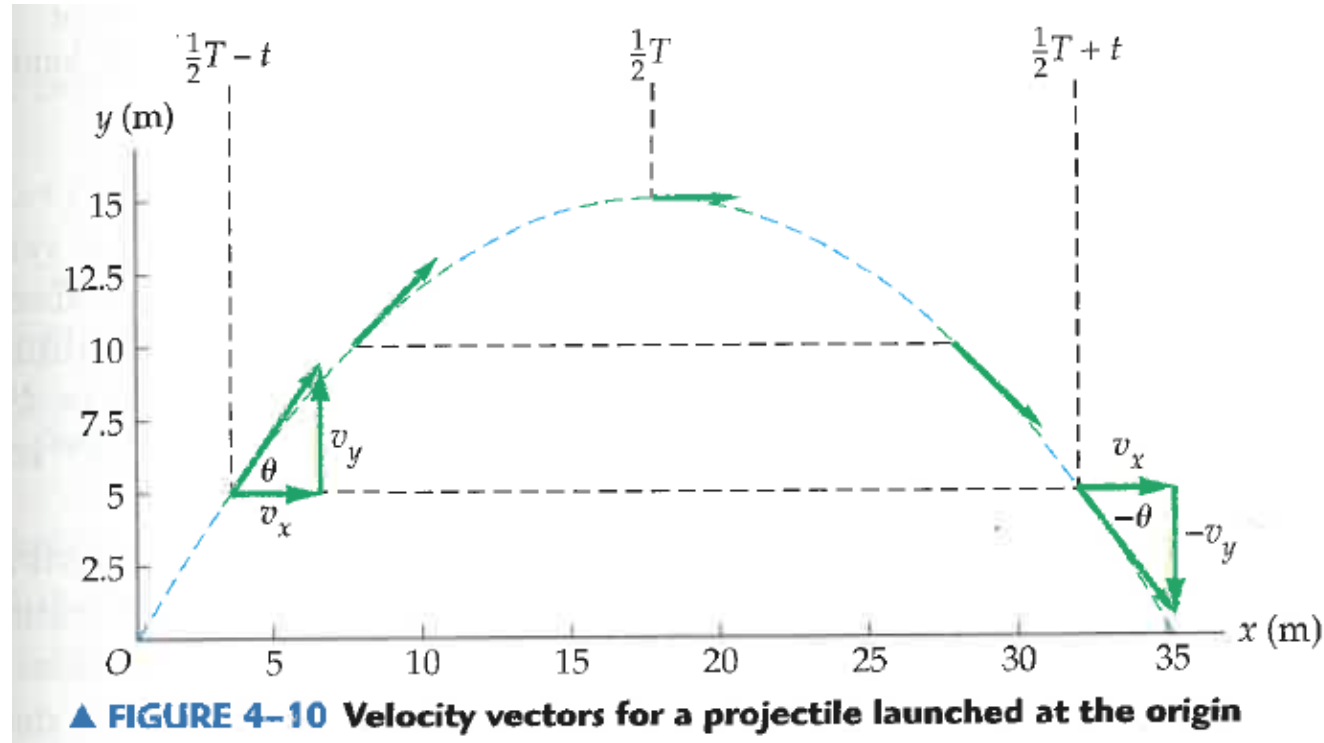


J. S. Walker, Physics, p87

https://phys.libretexts.org/Bookshelves/University_Physics/Physics_%28Boundless%29/3%3A_Two-Dimensional_Kinematics/3.3%3A_Projectile_Motion

Symmetry in projectile motion

From the parabolic trajectory, we can observe symmetry in projectile motion.



J. S. Walker, Physics, p94

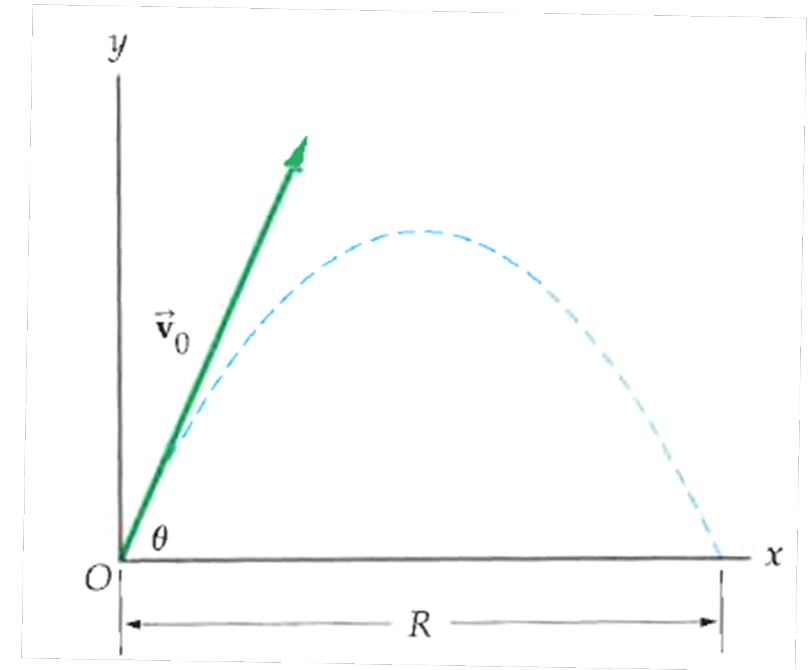
Symmetry in projectile motion

Find the time t_T when a projectile lands, where the initial and final elevations are the same.

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2$$

$$0 = v_{y,0}t_T - \frac{1}{2}gt_T^2$$

$$t_T = \frac{2v_{y,0}}{g}$$



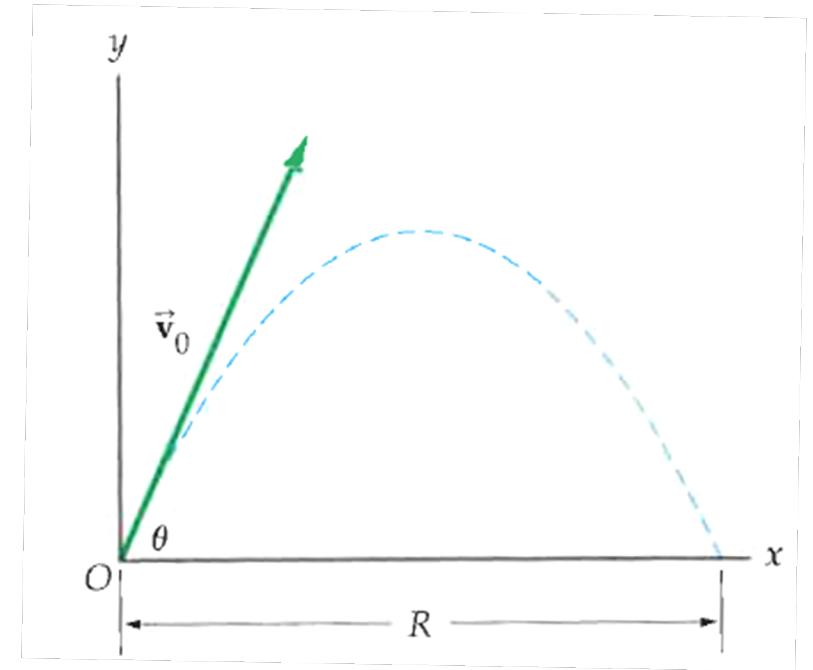
Symmetry in projectile motion

Find the time t_m when a projectile reaches its highest point.

$$v_y(t) = v_{y,0} - gt$$

$$0 = v_{y,0} - gt_m$$

$$t_m = \frac{v_{y,0}}{g}$$



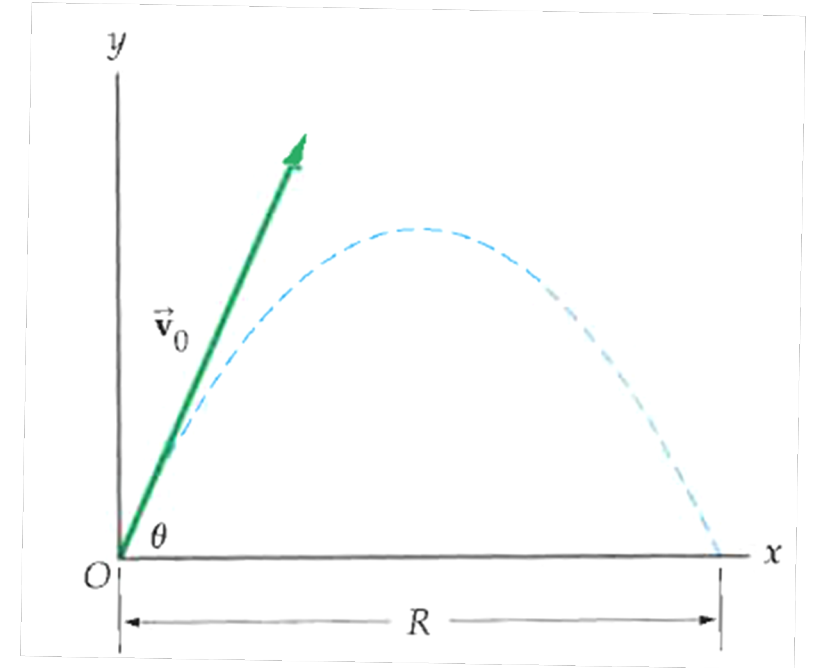
Symmetry in projectile motion

- The time (t_m) it takes a projectile to reach its highest point is one-half the time (t_T) it takes a projectile to land, where the initial and final elevations are the same.

$$t_m = \frac{1}{2} t_T$$

- Equivalently, the time (t_m) it takes a projectile to reach its highest point is the same as that ($t_T - t_m$) it takes from its highest point to landing point, where the initial and final elevations are the same.

$$t_m = t_T - t_m$$

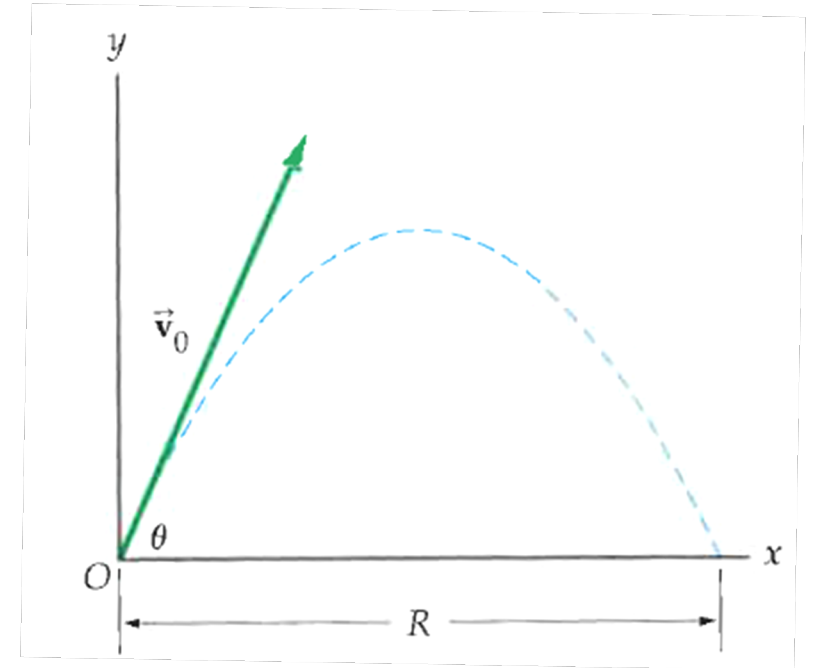


Symmetry in projectile motion

Find the velocity when a projectile lands, where the initial and final elevations are the same.

$$v_y(t) = v_{y,0} - gt$$

$$v_y(t_T) = v_{y,0} - gt_T = v_{y,0} - g \frac{2v_{y,0}}{g} = -v_{y,0}$$

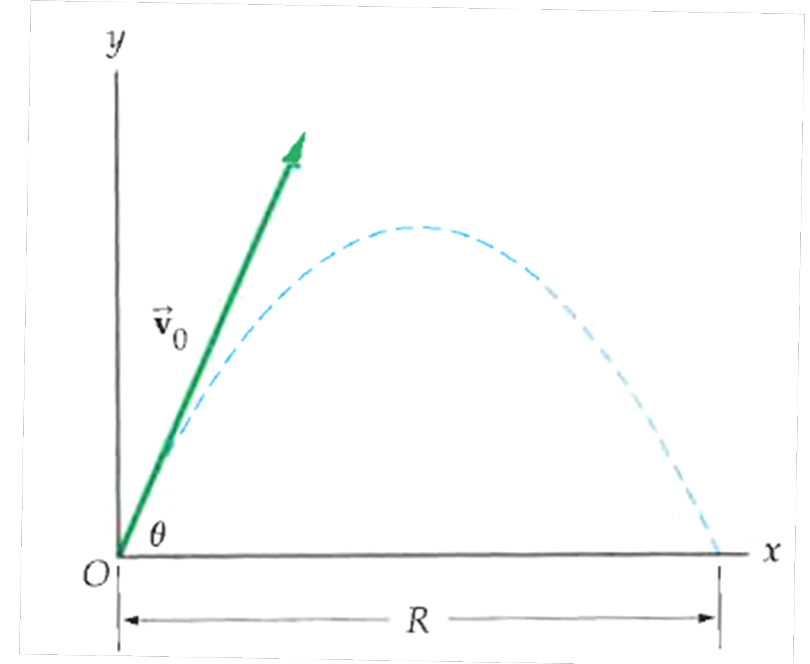


Symmetry in projectile motion

- The y component of the velocity when it lands is opposite to that when it was launched, where the initial and final elevations are the same.

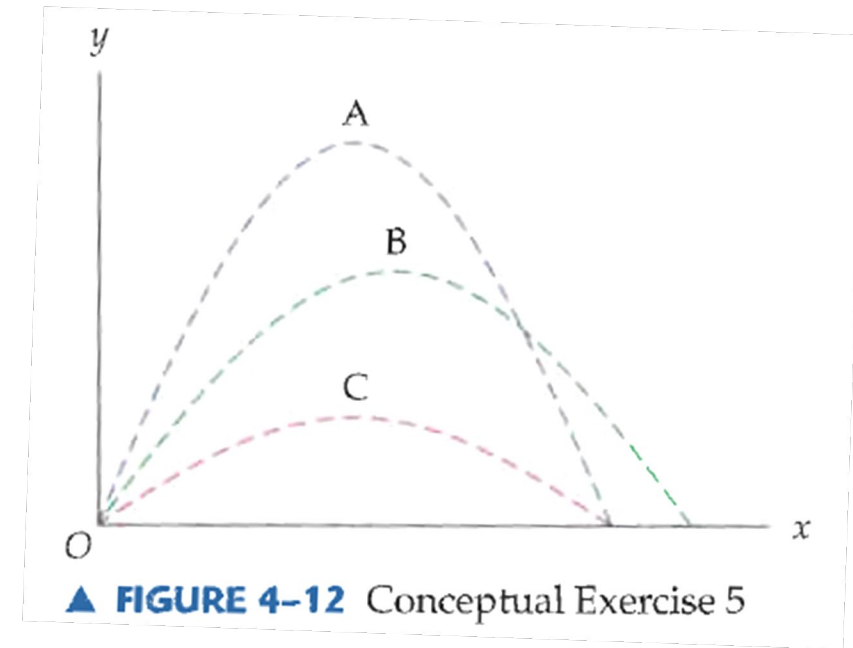
$$v_y(t_T) = -v_{y,0}$$

- The initial velocity is above the horizontal by an angle θ , and the landing velocity is below the horizontal by the same angle θ .
- The initial speed is the same as that when it lands.



Example

Three projectiles (A, B and C) are launched with the same initial speed but with different launch angles, as shown. List the projectiles in order of increasing time of flight. Indicate a tie with an equal sign.

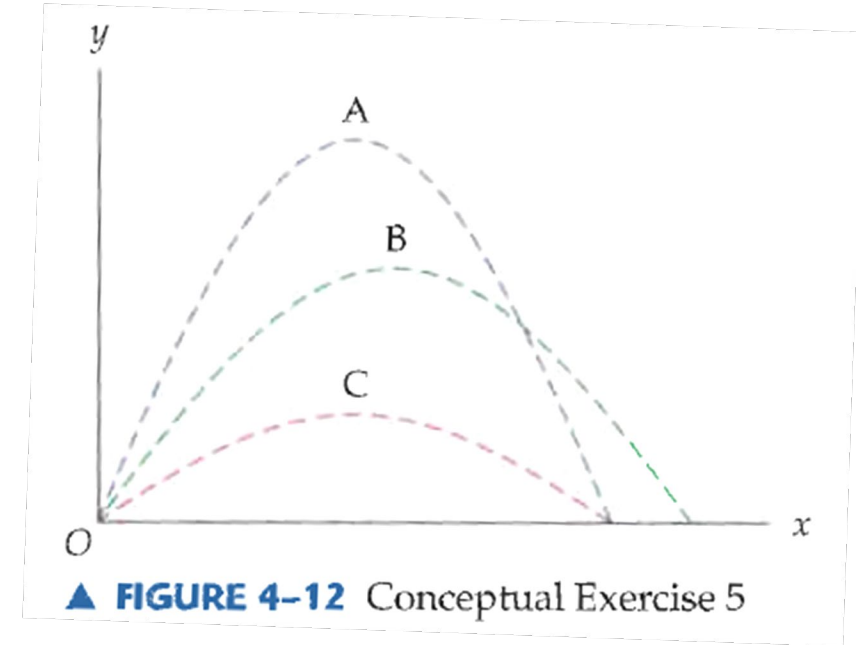


J. S. Walker, Conceptual Exercises 4-5, Physics, p100

Example: Solution

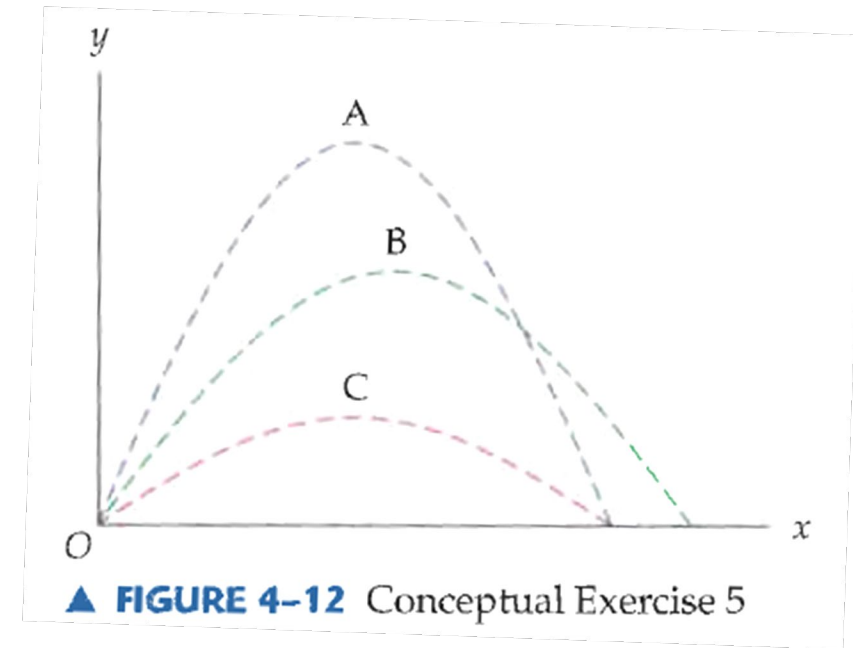
Three projectiles (A, B and C) are launched with the same initial speed but with different launch angles, as shown. List the projectiles in order of increasing time of flight. Indicate a tie with an equal sign.

Answer: C, B, A



Concept question 1

Three projectiles (A, B and C) are launched with the same initial speed but with different launch angles, as shown. List the projectiles in order of increasing time of reaching highest points. Indicate a tie with an equal sign.

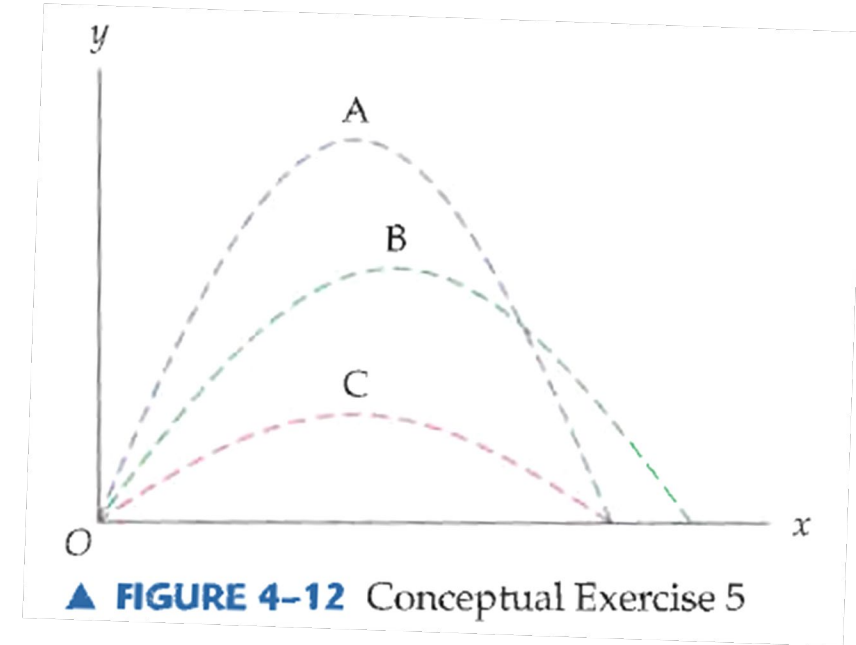


J. S. Walker, Conceptual Exercises 4-5, Physics, p100

Concept question 1

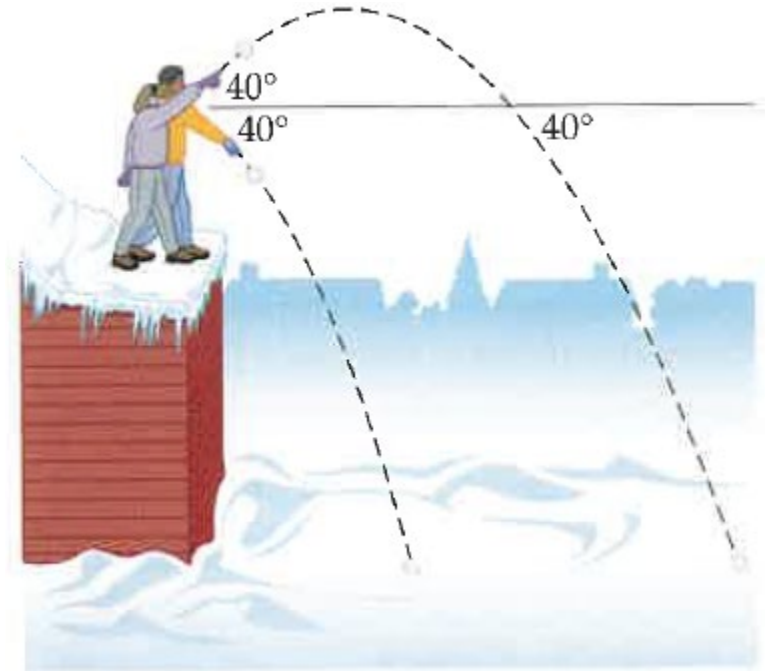
Three projectiles (A, B and C) are launched with the same initial speed but with different launch angles, as shown. List the projectiles in order of increasing time of reaching highest points. Indicate a tie with an equal sign.

Answer: C, B, A



Concept question 2

You and a friend stand on a snow-covered roof. You both throw snowballs with the same initial speed, but in different directions. You throw your snowball downward, at 40° below the horizontal; your friend throws their snowball upward, at 40° above the horizontal. When the snowballs land on the ground, is the speed of your snowball (a) greater than, (b) less than, or (c) the same as the speed of your friend's snowball?



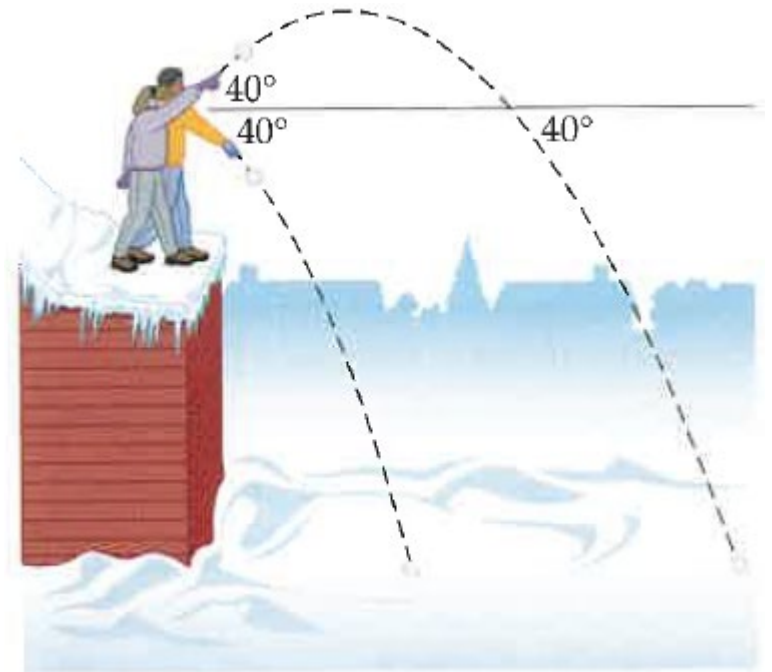
J. S. Walker, Conceptual Checkpoint 4-3, Physics, p95

Concept question 2: Solution

You and a friend stand on a snow-covered roof. You both throw snowballs with the same initial speed, but in different directions. You throw your snowball downward, at 40° below the horizontal; your friend throws their snowball upward, at 40° above the horizontal. When the snowballs land on the ground, is the speed of your snowball (a) greater than, (b) less than, or (c) the same as the speed of your friend's snowball?

Answer: (c)

J. S. Walker, Conceptual Checkpoint 4-3, Physics, p95



Example

A football game begins with a kickoff in which the ball travels a horizontal distance of 41m and lands on the ground. If the ball was kicked at an angle of 40.0° above the horizontal, what was its initial speed? Take $g = 9.81 \text{ m/s}^2$.

Example: Solution

A football game begins with a kickoff in which the ball travels a horizontal distance of 41m and lands on the ground. If the ball was kicked at an angle of 40.0° above the horizontal, what was its initial speed? Take $g = 9.81 \text{ m/s}^2$.

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2$$

$$0 = v_{y,0}t_T - \frac{1}{2}gt_T^2$$

$$t_T = \frac{2v_{y,0}}{g}$$

$$x = x_0 + v_{x,0}t$$

$$x(t_T) = v_{x,0}t_T = \frac{2v_{x,0}v_{y,0}}{g}$$

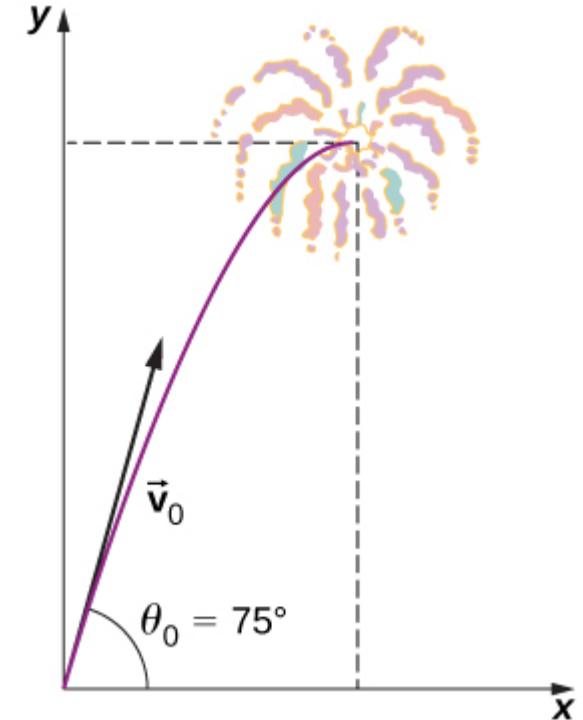
$$41 = \frac{2v_0 \cos \theta v_0 \sin \theta}{g}$$

$$v_0 = \sqrt{\frac{g \times 41}{2 \cos \theta \sin \theta}} = 20.2 \text{ m/s}$$

Case Problem 2

A shell is launched into air with an initial speed of 60.0 m/s at an angle of 75.0° above the horizontal. The shell is to explode when it reaches its highest point by timing the ignition. Take $g = 9.81 \text{ m/s}^2$.

- (a) Find the time of flight just before the explosion?
- (b) Find the highest height where the shell explodes.
- (c) Find the horizontal displacement of the shell when it explodes?
- (d) Find the position of the shell when it explodes.



<https://courses.lumenlearning.com/suny-osuniversityphysics/chapter/4-3-projectile-motion/>

Case Problem 2: Solution

A shell is launched into air with an initial speed of 60.0 m/s at an angle of 75.0° above the horizontal. The shell is to explode when it reaches its highest point by timing the ignition. Take $g = 9.81 \text{ m/s}^2$.

(a) Find the time of flight just before the explosion?

$$v_y(t) = v_{y,0} - gt$$

$$0 = v_{y,0} - gt_m \rightarrow t_m = \frac{v_{y,0}}{g} = \frac{v_0 \sin \theta}{g} = 5.9 \text{ s}$$

<https://courses.lumenlearning.com/suny-osuniversityphysics/chapter/4-3-projectile-motion/>

Case Problem 2: Solution

A shell is launched into air with an initial speed of 60.0 m/s at an angle of 75.0° above the horizontal. The shell is to explode when it reaches its highest point by timing the ignition. Take $g = 9.81 \text{ m/s}^2$.

(b) Find the highest height where the shell explodes.

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2$$

$$y(t_m) = y_0 + v_{y,0}t_m - \frac{1}{2}gt_m^2 = v_{y,0}t_m - \frac{1}{2}gt_m^2 = 171.2 \text{ m}$$

<https://courses.lumenlearning.com/suny-osuniversityphysics/chapter/4-3-projectile-motion/>

Case Problem 2: Solution

A shell is launched into air with an initial speed of 60.0 m/s at an angle of 75.0° above the horizontal. The shell is to explode when it reaches its highest point by timing the ignition. Take $g = 9.81 \text{ m/s}^2$.

(c) Find the horizontal displacement of the shell when it explodes?

$$x = x_0 + v_{x,0}t$$

$$x(t_m) = v_0 \cos \theta t_m = 91.6 \text{ m}$$

<https://courses.lumenlearning.com/suny-osuniversityphysics/chapter/4-3-projectile-motion/>

Case Problem 2: Solution

A shell is launched into air with an initial speed of 60.0 m/s at an angle of 75.0° above the horizontal. The shell is to explode when it reaches its highest point by timing the ignition. Take $g = 9.81 \text{ m/s}^2$.

(d) Find the position of the shell when it explodes.

$$\vec{r}(t) = x \hat{i} + y \hat{j}$$

$$\vec{r}(t) = 91.6m \hat{i} + 171.2m \hat{j}$$

<https://courses.lumenlearning.com/suny-osuniversityphysics/chapter/4-3-projectile-motion/>

Case Problem 3

You on the ground kick a ball into air at an angle of 53° above the horizontal. Its initial velocity is 10 m/s, find

- a) the maximum height achieved by the ball,
 - b) horizontal displacement when the ball lands on the ground, and
 - c) the total time required for the ball to land on the ground.
- ($\sin 53^\circ = 0.8$, and $\cos 53^\circ = 0.6$, Take $g = 9.81 \text{ m/s}^2$).

<https://brainly.in/question/15253680>

Case Problem 3: Solution

a) At highest point, $v_y = 0$

$$v_y = v_{y,0} - gt = 10 \sin 53^\circ - gt$$

$$v_y(t_m) = 10 \sin 53^\circ - gt_m = 0 \rightarrow t_m = 0.81 \text{ s}$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 = 0 + 10 \sin 53^\circ t - \frac{1}{2}gt^2$$

$$y(t_m) = 0 + 10 \sin 53^\circ t_m - \frac{1}{2}gt_m^2 = 3.3 \text{ m}$$

Case Problem 3: Solution

b) When the ball lands,

$$t_{total} = 2t_m$$

$$x = x_0 + v_{x,0}t$$

$$x(t_{total}) = x_0 + v_{x,0}t_{total} = 10 \cos 53^\circ t_{total} = 9.7 \text{ m}$$

c) When the ball lands,

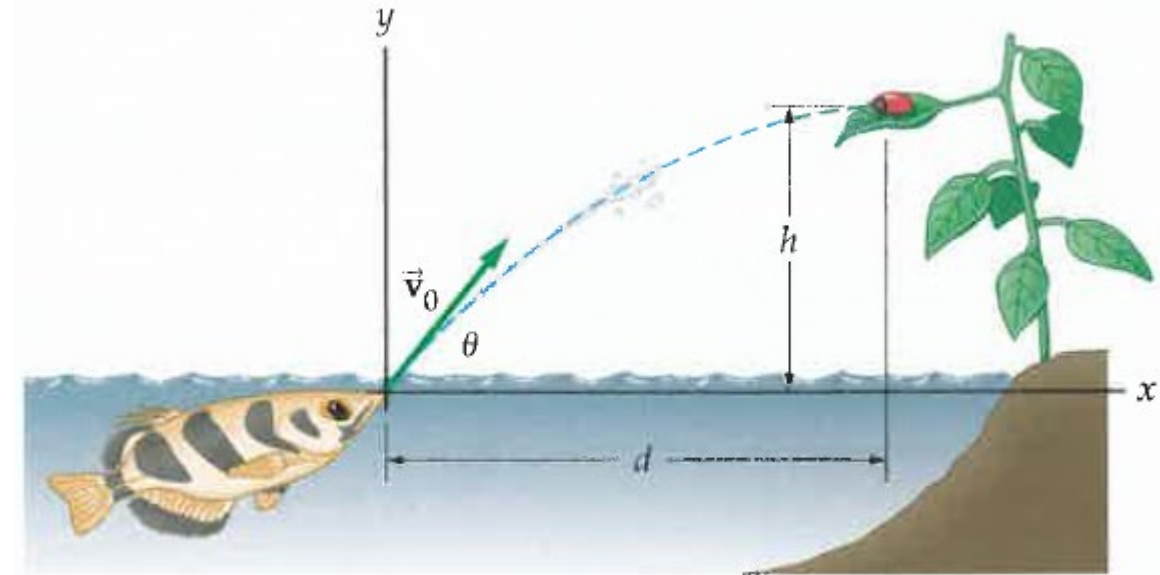
$$t_{total} = 2t_m = 1.62 \text{ s}$$

Off-class practice question(s)

Case Problem

The archerfish hunts by dislodging an unsuspecting insect from its resting place with a stream of water expelled from the fish's mouth. Suppose the archerfish squirts water with an initial speed of 2.30m/s at an angle of 19.5° above the horizontal. When the stream of water reaches a beetle on a leaf at height h above the water's surface, it is moving horizontally. Take $g = 9.81\text{ m/s}^2$.

- a) How much time does the beetle have to react?
- b) What is the height h of the beetle?
- c) What is the horizontal distance d between the fish and the beetle when the water is launched?



J. S. Walker, Physics, Example 4-7, p96

Case Problem: Solution

The archerfish hunts by dislodging an unsuspecting insect from its resting place with a stream of water expelled from the fish's mouth. Suppose the archerfish squirts water with an initial speed of 2.30m/s at an angle of 19.5° above the horizontal. When the stream of water reaches a beetle on a leaf at height h above the water's surface, it is moving horizontally. Take $g = 9.81\text{ m/s}^2$.

a) How much time does the beetle have to react?

$$v_y(t) = v_{y,0} - gt$$

$$0 = v_{y,0} - gt_m \rightarrow t_m = \frac{v_{y,0}}{g} = \frac{v_0 \sin \theta}{g} = 0.0783\text{ s}$$

Case Problem: Solution

The archerfish hunts by dislodging an unsuspecting insect from its resting place with a stream of water expelled from the fish's mouth. Suppose the archerfish squirts water with an initial speed of 2.30m/s at an angle of 19.5° above the horizontal. When the stream of water reaches a beetle on a leaf at height h above the water's surface, it is moving horizontally. Take $g = 9.81\text{ m/s}^2$.

b) What is the height h of the beetle?

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2$$

$$y(t_m) = y_0 + v_{y,0}t_m - \frac{1}{2}gt_m^2 = v_{y,0}t_m - \frac{1}{2}gt_m^2 = 0.03\text{ m}$$

Case Problem: Solution

The archerfish hunts by dislodging an unsuspecting insect from its resting place with a stream of water expelled from the fish's mouth. Suppose the archerfish squirts water with an initial speed of 2.30m/s at an angle of 19.5° above the horizontal. When the stream of water reaches a beetle on a leaf at height h above the water's surface, it is moving horizontally. Take $g = 9.81\text{ m/s}^2$.

- c) What is the horizontal distance d between the fish and the beetle when the water is launched?

$$x = x_0 + v_{x,0}t$$

$$x(t_m) = v_0 \cos \theta t_m = 0.17\text{ m}$$